

Assessing Infectiousness and the Impact of Effective Treatment to Guide Isolation Recommendations for People With Pulmonary Tuberculosis

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Determining the extent and duration of infectiousness of people with pulmonary tuberculosis (PWPTB) is critical for various aspects of tuberculosis care, including decisions regarding isolation. Studies suggest considerable heterogeneity in infectiousness of PWPTB. Pretreatment, measures of bacillary burden, including sputum smear microscopy, culture time to positivity, and Xpert MTB/RIF cycle threshold (Ct) value, predict the risk of transmission to contacts. Index patients with smear-negative disease pose lower infectious risk than those who have smear-positive disease, and household contact infection is more likely with index patients who have lower Xpert Ct values. Newer tools that enable detection of *Mycobacterium tuberculosis* complex from cough aerosol sampling and face mask sampling may be better predictors of contact infection risk. Clinical factors such as cough strength and frequency, and presence of cavitation on chest imaging, may also assist with risk prediction. Posttreatment, smear and culture status are poor predictors of infectiousness. While the exact duration of infectiousness post-treatment initiation remains uncertain, data from human-to-guinea pig transmission studies and clinical studies suggest that effective treatment results in a rapid decline in infectiousness, irrespective of smear or culture conversion. This is largely supported by early bactericidal activity and transcriptomic studies, as well as cough aerosol sampling studies, although a subset of patients may have persistent cough aerosol positivity. These findings can enable a more nuanced approach to isolation decision making while further research studies are awaited.

Keywords. tuberculosis; airborne infection; isolation; diagnostics; transmission.

Although most tuberculosis (TB) transmission occurs before diagnosis and effective treatment initiation, estimating the infectiousness of newly diagnosed patients impacts treatment decisions, the extent of contact tracing, and the duration of isolation. This translational science mini-review is based on a narrative evidence synthesis commissioned to inform the National Tuberculosis Coalition of America *Guidelines for Respiratory Isolation and Restrictions to Reduce Transmission of Pulmonary Tuberculosis in Community Settings* [1]. We examined the evidence supporting clinical factors that favor disease transmission, tests that can assess infectiousness, and the impact of effective treatment on transmission.

EXISTING AND NOVEL TESTS WITH IMPLICATIONS FOR THE ASSESSMENT OF INFECTIOUSNESS AT THE TIME OF DIAGNOSIS (TABLE 1)

Sputum Acid-Fast Bacilli Smear Microscopy

Despite its diagnostic accuracy limitations (a positive smear requires the presence of 5000–10 000 acid-fast bacilli/mL), sputum smear microscopy currently remains the most widely used measure of *Mycobacterium tuberculosis* complex (*Mtb*) bacillary burden [23, 24]. Cohort studies show that smear positivity predicts transmission risk prior to treatment initiation. A prospective cohort of 1004 patients with pulmonary TB (PWPTB) in the United Kingdom between 1948 and 1950 demonstrated that contact TB infection rates, measured using tuberculin skin test (TST) reactivity, varied according to index patient smear status: 65% (smear-positive, culture-positive), 27% (smear-negative, culture-positive), and 18% (smear-negative, culture-negative), compared with 22% in noncontacts [2]. The risk of TB disease was also higher in adult and child contacts exposed to index patients with smear-positive, culture-positive disease (11% and 14%) compared to smear-negative, culture-positive disease (0.07% and 3%) [2]. A retrospective cohort study from the Netherlands between 1967 and 1969 demonstrated that only 28% of smear-positive index

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Table 1. Existing and Novel Tests With Implications for the Assessment of Infectiousness at the Time of Diagnosis

Study Type and Year [Reference]	Population and Setting	Key Findings
Sputum AFB smear positivity pretreatment increases transmission risk		
Shaw & Wynn-Williams, 1953 [2] Prospective cohort	1004 PTB, 823 child contacts; UK	TST reactivity in contacts varied by index patient smear status: 65% (smear-positive index patients), 27% (smear-negative, culture-positive index patients), 18% (smear- and culture-negative index patients), compared to 22% (group without TB contact exposure). TB disease risk for adult and child contacts was higher if exposed to smear-positive index patients (11% and 14%) vs smear-negative, culture-positive patients (0.07% and 3%).
van Geuns et al, 1975 [3] Retrospective cohort	299 PTB, 2817 contacts with 478 child contacts of smear-positive cases; The Netherlands	TST conversion occurred in 50% of household contacts of smear-positive index patients compared to 6% of household contacts of smear-negative index patients.
Behr et al, 1999 [4] Molecular epidemiology	1574 culture-positive PTB in 71 clusters, 183 secondary cases; US	Relative rate of transmission from smear-negative patients: 0.22 (95% CI, .16–.32).
Tostmann et al, 2008 [5] Molecular epidemiology	1285 PTB in 394 clusters, 844 secondary cases; The Netherlands	Relative rate of transmission from smear-negative patients: 0.24 (95% CI, .20–.30).
Asadi et al, 2022 [6] Genomic epidemiology	1176 PTB cases, 21 131 contacts; Canada	Relative rate of transmission from smear-negative patients: 0.10 (95% CI, .05–.19); 8% transmission (95% CI, 3%–14%).
NAATs: Xpert Ct values correlate with bacillary burden and predict transmission to contacts		
Lange et al, 2017 [7] Meta-analysis	7511 samples from 4447 PTB; Various	Xpert Ct values are strongly associated with smear status, with an area under the curve of 0.85 (95% CI, .82–.87).
Namugenyi et al, 2021 [8] Prospective cohort	138 PWH and PTB; Uganda	For each unit increase in baseline Ct, colony grade was reduced: 0.86 times (OR, 0.86 [95% CI, .74–.97]; week 2) and 0.84 times (OR, 0.84 [95% CI, .79–.95]; $P = .002$, week 4). Each unit increase in Ct was associated with a 3% higher likelihood of earlier conversion to negativity.
Najjingo et al, 2019 [9] Cross-sectional	495 PWPTB; regional referral hospitals in Uganda	Xpert Ct values were found to increase by 2.57 for every unit increase in days to positivity. HIV was associated with high Xpert Ct values and days to positivity.
Hanrahan et al, 2014 [10] Cross-sectional	2406 PWPTB, paired sputum samples; South Africa	PWH had higher Xpert Ct values compared to those without HIV (26.6 vs 22.9; $P < .001$). Higher average Xpert Ct values if CD4 count was <200 cells/ μ L (3.66 cycles; $P < .001$).
Garcia et al, 2024 [11] Prospective observational	382 PWPTB, 1055 close contacts; Brazil	Lower baseline Xpert Ct values in index patients were associated with increased risk of QuantiFERON-Plus conversion in contacts (OR, 1.61 [95% CI, 1.12–2.32]).
CASS: The majority of PWPTB do not produce culturable <i>Mtb</i> from aerosols, but higher numbers of <i>Mtb</i> CFUs from cough aerosols are a better predictor of transmission risk than sputum smear microscopy		
Fennelly et al, 2004 [12] Prospective cohort	16 smear-positive adults; US	25% of PTB were CASS positive; increased cough frequency associated with CASS positivity ($P = .008$).
Fennelly et al, 2012 [13] Prospective cohort	101 people with culture-positive PTB; Uganda	28% of PTB were CASS positive; positive aerosol cultures were associated with stronger cough ($P = .016$), higher sputum smear microscopy grade ($P = .007$), and lower days to positive in liquid culture ($P = .004$).
Jones-López et al, 2013 [14] Prospective cohort	96 clinic patients with sputum culture-positive PTB and 442 contacts; Uganda	Exposure to a high-aerosol-producing index patient (≥ 10 CFUs) was the only predictor of contact incident infection (OR, 4.81 [95% CI, 1.20–19.23]).
Jones-López et al, 2016 [15] Retrospective cohort	85 patients with smear-positive TB on treatment for <5 d and 369 household contacts; Uganda	Incident TB disease in contacts was associated with exposure to index patient with high level of cough aerosols (OR, 6 [95% CI, 1.4–25.2]; $P = .01$).
Acuña-Villaorduña et al, 2018 [16] Cross-sectional	48 smear- and culture-positive index PTB patients with 230 household contacts; 16 TB clinics in Brazil	IGRA levels were higher in contacts of index patients producing higher cough aerosols (median, 4.6 IU/mL [IQR, 0.02–10]) compared to low cough aerosols (0.8 IU/mL [0.2–10]) and no cough aerosols (0.1 IU/mL [0–3.7]; $P = .08$). 89% (8/9) of culture positive secondary cases occurred in contacts of index patients producing cough aerosols.
Acuña-Villaorduña et al, 2019 [17] Aggregate analysis	233 culture-positive PTB patients; Uganda and Brazil	Only 100 (43%) had culturable <i>Mtb</i> in cough aerosols (18% low aerosol burden, 1–9 CFUs; 24% high burden, >10 CFUs). Fewer days of TB treatment (OR, 1.47 per 1 d of treatment [95% CI, 1.16–1.89]; $P = .001$) associated with aerosol positivity. 30%–40% of PWPTB who produce cough aerosols rapidly become nonculturable after treatment initiation.
Theron et al, 2020 [18] Cross-sectional	500 people with TB (including 227 with DR-TB); South Africa	31% were CASS-positive, of whom 30% had ≥ 10 CFUs cultured from aerosols Clinical factors (fewer symptoms, stronger cough, no HIV, treatment ≤ 48 h) and mycobacterial factors (higher sputum bacillary load based on smear microscopy grade and TTP) independently predicted the presence of culturable <i>Mtb</i> bacilli in cough aerosol.

Table 1. Continued

Study Type and Year [Reference]	Population and Setting	Key Findings
Face mask sampling: Detection of culturable <i>Mtb</i> from FMS does not correlate with other measures of bacillary burden, and higher level of <i>Mtb</i> detectable from FMS was associated with a higher risk of transmission		
Williams et al, 2014 [19] Longitudinal prospective	17 people with smear-positive PTB in series 1, 29 smear-positive PTB in series 2; UK and The Gambia	Exhaled <i>Mtb</i> output did not have a diurnal pattern and was not associated with cough frequency, sputum bacillary content, or chest radiographic disease severity.
Williams et al, 2020 [20] Longitudinal prospective	Longitudinal: 24 people with microbiologically confirmed PTB; South Africa Active case finding pilot study: 20 people with TB symptoms; South Africa	<i>Mtb</i> detected in 86% using FMS and 21% from sputum samples during a 24-h period. No association between <i>Mtb</i> output and bacillary burden (based on smear status, Xpert grade, or TTP) or radiological findings. Of 8 participants diagnosed with PTB, 6 were exclusively positive by FMS, 4 of whom had treatment responsive early TB-compatible lesions on PET-CT and developed Xpert-positive samples at 6 wk.
Williams et al, 2023 [21] Prospective cohort	46 people with smear-positive PTB; The Gambia	91% of patients with smear-positive PTB had detectable <i>Mtb</i> on FMS with high variation in IS6110 copy numbers. High face mask <i>Mtb</i> levels ($\geq 200\,000$ IS6110 copies) independently associated with higher likelihood of incident TB infection measured by QuantiFERON conversion (OR, 3.2 [95% CI, 1.26–8.12]).
Hassane-Harouna et al, 2023 [22] Prospective	150 PWPTB, Guinea	Lower sensitivity (48%) for FMS to detect <i>Mtb</i> in patients with a positive sputum Xpert MTB/RIF, although specificity was high (96%).
Abbreviations: AFB, acid-fast bacilli; CASS, cough aerosol sampling systems; CFU, colony-forming units; CI, confidence interval; Ct, cycle threshold; DR-TB, drug-resistant tuberculosis; FMS, face mask sampling; HIV, human immunodeficiency virus; IGRA, interferon-gamma release assay; IQR, interquartile range; <i>Mtb</i> , <i>Mycobacterium tuberculosis</i> ; NAAT, nucleic acid amplification test; OR, odds ratio; PET-CT, positron emission tomography-computed tomography; PTB, pulmonary tuberculosis; PWH, people with human immunodeficiency virus; PWPTB, people with pulmonary tuberculosis; TB, tuberculosis; TST, tuberculin skin test; TTP, time to culture positivity; UK, United Kingdom; US, United States.		

patients infected any of their contacts [3]. TST conversion occurred in 50% of child contacts if the index patient was smear-positive compared to 6% if smear-negative [3]. Molecular fingerprinting studies show that only 13%–17% of transmission arises from sputum smear-negative TB (relative transmission rate of 0.19–0.24) compared to smear-positive [4, 5]. More recent genomic epidemiology analyses suggest that an even lower proportion of transmission (8%) may be attributable to persons with smear-negative TB [6].

Time to Culture Positivity

Time to culture positivity (TTP) is a measure of metabolic activity of viable *Mtb* in liquid sputum cultures, which correlates with the number of colony-forming units (CFUs) of *Mtb* on solid cultures [25]. Shorter TTP (<9 days vs ≥ 9 days), which correlates with a higher radiographic disease burden, has been associated with increased rates of contact infection or disease (odds ratio [OR], 2.10 [95% confidence interval {CI}, 1.37–3.24]; $P = .001$) [26]. Other tools such as TB quantitative viability microscopy have also been shown to predict quantitative culture results and treatment responses, suggesting a potential role in assessing infectiousness [27, 28].

Nucleic Acid Amplification Tests

Molecular World Health Organization–recommended nucleic acid amplification tests (NAATs) are recommended as the initial diagnostic test for all persons being evaluated for TB. Xpert MTB/RIF (GeneXpert, Cepheid Inc, Sunnyvale, California) provides semi-quantitative results in the form of the cycle threshold (Ct), which denotes the number of DNA amplification cycles required for *Mtb* DNA to become detectable. An

individual-level data meta-analysis (7511 samples from 4447 patients) demonstrated that Xpert Ct values are strongly associated with smear status, with an area under the curve of 0.85 (95% CI, .82–.87) [7]. Higher Ct values indicate a lower mycobacterial burden: As baseline Ct values increase, TTP increases and CFUs decrease [8–10]. A multicenter Brazilian cohort study demonstrated that lower baseline Xpert Ct values in index PWPTB (80% of whom had smear-positive disease) were associated with increased risk of interferon-gamma release assay (IGRA) conversion in close contacts (OR, 1.61 [95% CI, 1.12–2.32]) [11]. Further studies are needed to evaluate the correlation between Xpert Ultra Ct values and contact infection, including in people with human immunodeficiency virus (HIV) (who may have higher Ct values), and using different thresholds for IGRA positivity [29].

Detection of Culturable *Mtb* From Bioaerosol Sampling

Beyond the presence of smear or culture positivity, the presence of an effective cough to generate aerosol appears to be an important predictor of transmission. Cough aerosol sampling systems (CASS) enable direct measurement of aerosolized culturable *Mtb* bacilli using Anderson 6-stage cascade samplers that enable viable CFUs from individual aerosol droplets to be isolated in a size-dependent manner [12, 18]. Importantly, a minority of PWPTB produce culturable aerosols (estimates between 31% and 45%), and only 30%–50% of these produce >10 CFUs [14, 15, 18]. The detection of culturable cough aerosols is associated with stronger and more frequent coughing [12, 13, 15, 18]. Cough aerosol *Mtb* detection (currently only available as a research tool) appears to be a better predictor of infectiousness than sputum smear microscopy [14]. Higher numbers of

Mtb CFUs in cough aerosols and a more severe cough in index PWPTB are associated with higher transmission risk, measured by progression to TB disease and/or the development of TB infection (TST or IGRA conversion) in contacts [14, 15, 30]. High transmission household contacts of cough aerosol-positive index cases had stronger IGRA responses [14, 16].

While some CASS studies identified no significant association between culturable cough aerosol detection, sputum smear grade, and TTP [12, 14, 16], 1 study only enrolled 16 participants (proof-of concept study) [12], another only included 2+/3+ smear grades [16], and the aggregate analysis of Ugandan and Brazilian data was underpowered to detect an association with smear grade [14]. Several studies found that higher smear grade was associated with CASS positivity [13, 15, 18] and with increased CFUs detected from aerosols [15, 17]. More recent data demonstrate that CASS positivity is associated with smear grade [18, 31], cavitation, and strong cough [31], and that contacts of CASS-positive index patients had higher quantitative IGRA results [31], suggesting an inoculum effect that reflects infectious dose [32]. The high variability in cough aerosol culture positivity (which may reflect a combination of infectious dose, *Mtb* genetics, and host characteristics) suggests there is likely to be variability in the amount of transmission mediated by cough, independent of smear or culture results [12–14].

Viable *Mtb* has been detected in aerosols generated by participants sampled during tidal breathing (without cough) using a respiratory aerosol sampling chamber (RASC) that uses culture-independent methods to measure live cell fluorescence of bacilli arrayed in a Nanowell device [33–35]. The high proportion of participants from whom viable *Mtb* was isolated (>90% of those with both positive and negative sputum Xpert Ultra results) raises questions about the importance of cough for transmission. Similarly, the finding that viable *Mtb* was isolated from 20%–30% of participants after 6 months of TB treatment highlights the possibility of prolonged infectiousness in a subset of patients. However, since infectiousness declines prior to smear and culture conversion [36], viable or culturable *Mtb* does not necessarily equate to infectious *Mtb*. Since RASC studies did not directly measure transmission, the clinical and public health implications remain uncertain.

Detection of Exhaled *Mtb* Using Face Mask Sampling

Another approach for quantifying exhaled *Mtb* bacilli involves PWPTB wearing modified face masks containing a gelatin filter that is subsequently dissolved, heated, and centrifuged to generate a pellet that can be analyzed using the Xpert platform [19, 20]. A study in South Africa demonstrated that exhaled *Mtb* output detected by face mask sampling (FMS) in 24 PWPTB did not have a diurnal pattern and was not associated with cough frequency, sputum bacillary load, or chest

radiographic disease severity [19]. In a prospective study of 20 patients with TB symptoms in South Africa, of the 8 diagnosed with pulmonary TB, 6 were exclusively positive on FMS but not on initial sputum Xpert, of whom 4 subsequently developed Xpert-positive sputum specimens 6 weeks later [20]. An FMS study conducted in The Gambia demonstrated that 91% of smear-positive PWPTB had detectable *Mtb* on FMS with high variation in IS6110 copy numbers [21]. High face mask *Mtb* levels ($\geq 20\,000$ IS6110 copies), seen in 45% of patients, were independently associated with a higher likelihood of incident TB infection (IGRA conversion: OR, 3.2 [95% CI, 1.26–8.12]) [21]. The correlation between the detection of culturable exhaled *Mtb* (86%) from FMS and sputum smear positivity is poor (21%) [17, 30]. Efforts to develop FMS prototypes are under way; however, a recent study from Guinea demonstrated a lower sensitivity (48%) for 1 such FMS prototype to detect *Mtb* in patients with a positive sputum Xpert MTB/RIF, although specificity was high (96%) [22].

CLINICAL FACTORS THAT AFFECT INFECTIOUSNESS AND TRANSMISSION RISK (TABLE 2)

Cough Dynamics

Cough, often measured in terms of frequency [44] or intensity [30], is a clinical predictor of infectiousness. This is supported by studies showing that contacts of index patients with cough compared to no cough [37, 38, 45], and higher cough frequency, are more likely to be infected [41, 46], but other studies found no significant association [39, 40]. These data are challenging to interpret since study design and cough assessments varied, including retrospective record review [38], evaluation of a symptomatic prospective cohort [39], and use of a more precise cough monitor [41, 42, 47]. Cough frequency is associated with cavities that are larger or closer to the airways, and cough in treated patients ceased 3-fold faster with smaller cavities and cavities located farther away from the airway [42, 47]. Cough severity, which was measured with 2 cough questionnaires (a visual analogue cough scale and the Leicester Cough Questionnaire), has also been associated with high-transmission households, where >70% of contacts had TST conversion >10 mm [30]. The relationship between cough and transmission is complex. There are likely differences between voluntary cough (measured in CASS studies and known to increase *Mtb* output) and involuntary or spontaneous cough (often incessant due to airway irritation, particularly in the setting of cavitation), but this is not well studied.

Cavitary Lung Disease

Pulmonary cavities, driven by liquefaction necrosis, serve as a niche within the infected lung parenchyma that has a particularly high bacillary load [48] and may be resistant to the penetration of anti-TB medication [49], which can increase or

Table 2. Clinical Factors That Affect Infectiousness and Transmission Risk

Study Type and Year	Population and Setting	Key Findings
Cough in index patients is associated with bacillary burden and increased transmission risk		
Gessner et al, 1998 [37] Case-control	146 PWPTB with 302 household contacts <15 years; Alaska	Children more likely to develop TB infection if exposed to adult index patients who coughed (OR, 2.8 [95% CI, 1.3–6.1]).
Lohmann et al, 2012 [38] Retrospective	255 index TB PWPTB, 3509 TST-screened contacts, 2679 CXR-screened contacts; Dutch Municipal Health Service	Cough was a risk factor for PTB transmission (RR, 4.2 [95% CI, 2.10–8.56]). Cavitation was also a risk factor for transmission (RR, 4.2 [95% CI, 2.10–8.56]).
Otero et al, 2016 [39] Longitudinal prospective	1178 smear-positive PWPTB, 5466 household contacts; Lima, Peru	Index patients without cough were less likely to transmit PTB to contacts (crude OR, 0.5 [95% CI, .1–.5]). High bacillary load was associated with transmission (IRR, 1.5 [95% CI, 1.02–2.3]).
Elmi et al, 2014 [40] Cross-sectional	69 PWPTB, 70 household contacts; Malaysia	Cough presence was associated with TB infection in contacts (aOR, 0.15 [95% CI, .03–.62]; $P < .009$).
Turner et al, 2018 [41] Cross-sectional	44 PWPTB, 17 LTBI; United Kingdom	Cough frequency was associated with TB infection in contacts (OR, 1.30 [95% CI, 1.04–1.64]; $P = .022$).
Proaño et al, 2018 [42] Prospective cohort	41 PWPTB: HIV-negative, culture-positive DS-TB; Lima, Peru	Cough frequency was associated with larger cavities on chest imaging and cavities closer to the airways, and cough ceased 3 times faster with smaller cavities further away from the airway.
Cavitation in index patients is associated with bacillary burden and increased transmission risk		
Jones-López et al, 2014 [30] Prospective cohort	124 smear-positive PWPTB, 731 contacts; network of 5 laboratories serving 16 clinics in Brazil	Unadjusted univariate model suggested cavitation was associated with transmission (OR, 3.13 [95% CI, 1.13–8.69]; $P = .028$); trended toward significance in adjusted analyses ($P = .078$).
Proaño et al, 2018 [42] Prospective cohort	41 PWPTB: HIV-negative, culture-positive DS-TB; Lima, Peru	Cough frequency during treatment was higher in participants with large cavity volumes (rate ratio, 1.98; $P = .01$) and cavities located closer to the airway (rate ratio, 2.44; $P = .001$).
Melsew et al, 2018 [43] Meta-analysis	Systematic review including 37 studies	Contacts of patients were 2 times as likely to get infected by patients with cavitory disease than those without cavitory disease (pooled aOR, 1.9 [95% CI, 1.26–2.84]; $I^2 = 63\%$).
Garcia et al, 2024 [11] Prospective observational	382 PWPTB, 1055 close contacts; Brazil	Contacts who developed TB infection (IGRA conversion) were more likely to have been exposed to an index patient with cavitation on CXR in contrast to IGRA-negative contacts (54.7% vs 40.2%; $P = .007$).

Abbreviations: aOR, adjusted odds ratio; CI, confidence interval; CXR, chest X-ray; DS-TB, drug-susceptible tuberculosis; HIV, human immunodeficiency virus; IGRA, interferon-gamma release assay; IRR, incidence rate ratio; LTBI, latent tuberculosis infection; OR, odds ratio; PWPTB, people with pulmonary tuberculosis; RR, risk ratio; TB, tuberculosis; TST, tuberculin skin test.

prolong transmission risk. People with cavitory disease are more likely to have smear-positive disease [18, 50] and have higher relapse rates [51–53]. Cavities promote coughing, which could in turn lead to transmission [43]. Contact tracing studies demonstrate the link between cavitory lesions and higher risk of TB transmission to contacts [30], which has been corroborated in a meta-analysis [43]. The Brazilian cohort study evaluating the association between Xpert Ct values and transmission showed that cavitory lesions were also an independent predictor of transmission risk (OR, 2.32 [95% CI, 1.3–4.13]) [11].

Superspreading Events

Superspreading events are likely due to a combination of source, pathogen, host, and environmental factors, leading to index patients accounting for multiple secondary TB infections [54]. The scale-up of whole genome sequencing [55] has overcome challenges in linking secondary infections to index patients due to long latency periods [56]. A contact tracing study in Australia reported that 72.5% (2415/3213) of all secondary infections were linked to superspreading events,

defined as causing as many or more secondary infections as the 99th centile of an equivalent homogeneous distribution [54, 57]. A systematic review investigating TB transmission dynamics estimated that the minority of PWPTB (2%–31%) drive the majority (80%) of transmission at a population level [58]. Yet while multicountry datasets showed that 20% of patients account for 50% of total cough days (supercoughers), characteristics predictive of such individuals were not identified, highlighting the challenge of identifying patients with higher ongoing transmission risk [56].

BACTERIAL DETERMINANTS OF TRANSMISSIBILITY

While this review focuses primarily on host and clinical factors, including the effect of treatment on infectiousness, we note that intrinsic bacterial characteristics of *Mtb* also facilitate transmission. A household contact study proposed distinct strains may cause different levels of cavitation and cough severity, leading to different transmission patterns [30]. More recent mouse model studies found that high-transmission strains of *Mtb* generated different immunopathologies compared to low-transmission strains [59, 60]. High-transmission strains induced granulomas

with a greater potential to develop into lung cavities, which were more likely to lead to caseous necrosis, whereas low-transmission strains generated diffuse inflammatory lesions [59, 60].

EFFECT OF TREATMENT ON TRANSMISSION RISK (TABLE 3)

Prompt initiation of effective treatment for both drug-susceptible TB (DS-TB) and drug-resistant TB (DR-TB) is critical to improve individual outcomes and to prevent transmission [74–76]. Effective treatment is defined as a multidrug regimen to which the organism is susceptible or anticipated to be susceptible, ideally with initial rapid molecular drug susceptibility testing (DST), subsequently confirmed with culture-based phenotypic and/or genotypic DST [75]. However, the duration of effective treatment required to decrease transmission remains uncertain and controversial [77, 78]. The 2-week isolation recommendation most commonly applied across settings was derived from a 1976 review, which stated that “abolition of the patient’s infectiousness is very probably obtained after <2 weeks of treatment” [36]. Data leveraging a range of study designs and experimental methodologies, including in vitro studies, human-to-guinea pig (H-GP) transmission models, transcriptomics, and clinical cohorts, suggest that effective treatment leads to a rapid decline in TB transmission. Importantly, most studies evaluated the duration of infectiousness at early points in time (24–72 hours) following treatment initiation versus longer durations (eg, 2 weeks); however, interval durations have not been systematically evaluated.

Early Bactericidal Activity Studies

Early bactericidal activity (EBA) studies demonstrated a rapid decrease in CFUs/mL of sputum per day over the first 2 days, followed by a slower decrease over the following 12 days with combination regimens including isoniazid, rifampin, pyrazinamide, ethambutol, and streptomycin, with differences noted between drugs and doses of drugs [61, 62]. In the first 2 days, isoniazid has the greatest EBA, followed by rifampin, whereas from days 2 to 14, rifampin has significant bactericidal effects, highlighting its important sterilizing activity [62].

Culture-Positive Cough Aerosols Are Higher Pretreatment and Decrease After Treatment Initiation

An early study of patients with smear- and culture-positive TB in the United States showed a rapid decrease in culturable *Mtb* from cough aerosols within the first 3 weeks of TB treatment, and CASS positivity was associated with lack of TB treatment during the prior week [12]. Univariate analyses from a study in Uganda found an association between fewer days of TB treatment and CASS positivity [13]. In an aggregate analysis from three cohorts of PWPTB in Uganda and Brazil, of the 43% producing culturable *Mtb* from cough aerosols, only shorter duration of treatment before CASS was associated with aerosol positivity (OR, 1.47 [95% CI, 1.16–

1.89] per day of treatment; $P = .001$). This remained statistically significant when the analysis was limited to patients with high aerosol CFU counts (OR, 1.38 [95% CI, 1.05–1.82]; $P = .02$) [17]. There was a dose-response effect: when compared with patients who received >2 days of TB treatment, the OR for aerosol positivity was 2.56 (95% CI, 1.11–5.92; $P = .03$) for those who received 1–2 days of treatment, and 5.5 (2.3–13.4; $P < .001$) for those not on treatment [17]. Similarly, other studies showed cough frequency reduced with treatment [42, 46], suggesting that including cough assessment may improve transmission prediction.

In a cohort of 500 South African PWPTB, CASS positivity reduced from 52% and 38% of DS-TB and DR-TB patients, respectively, on <48 hours of treatment, to 34% and 21% after 48 hours of treatment [18]. CASS positivity was detected in a minority of DS-TB patients on 3, 4, 5, and 6 days of treatment prior to CASS (9/26 [35%], 11/31 [35%], 12/37 [32%], and 7/26 [27%], respectively) [18]. In analyses limited to patients on first-line or fluoroquinolone-based regimens, likely drug effectiveness was not associated with cough aerosol positivity [18]. When baseline CASS-positive PWPTB were resampled after months of treatment, 5% (3/66) with DS-TB and 40% (7/18) with DR-TB (albeit the number of likely effective drugs in this group was low) displayed prolonged CASS positivity [18].

Transcriptomic Studies Demonstrate Changes in *Mtb* Transcriptional Profiles That May Lower Pathogenicity and Infectiousness Within Days After Treatment Initiation

Gene expression studies evaluating *Mtb* isolates from patients with DS-TB in Uganda demonstrate that bacterial messenger RNA (mRNA) abundance declined >98% within 4 days, indicating rapid killing [63]. Another study in India used Illumina sequencing to assess *Mtb* mRNA transcriptional signatures at 1, 3, 5, 7, and 14 days of treatment [64]. Downregulation of genes associated with bacterial persistence, with specific drug-induced changes, occurred from day 3 onward [64]. Following exposure to sterilizing TB drugs, *Mtb* bacilli exhibit transcriptomic changes that may impact pathogenicity and virulence and occur as early as 1 day after treatment initiation, despite ongoing viability and detection in mycobacterial cultures [64].

Measurement of precursor ribosomal RNA (rRNA) demonstrates that sterilizing drugs, like bedaquiline and pyrazinamide, and regimens containing these drugs have a potent suppressive effect on *Mtb* rRNA synthesis, in contrast to nonsterilizing drugs like isoniazid or ethambutol, or weaker regimens [65]. While suppression of the rRNA synthesis ratio occurred within hours of exposure to bedaquiline in vitro, indicating near-cessation of rRNA synthesis, CFUs did not decrease substantially until 8–12 days, and results for pyrazinamide were similar [65]. These data highlight the potential value of measures like rRNA synthesis ratio, that are distinct from measures of bacterial burden, to assess infectiousness, since drugs and regimens impact these measures differently.

Table 3. Effect of Tuberculosis Treatment on Measures of *Mycobacterium tuberculosis* Complex Bacillary Burden and Transmission Risk

Study Type and Year	Population and Setting	Key Findings
Early bactericidal activity: TB drugs have a differential effect on EBA		
Jindani et al, 1980 [61] Prospective cohort	124 newly diagnosed, previously untreated PWPTB (smear-positive); Africa	A rapid decrease in CFU/mL of sputum per day over the first 2 days followed by a slower decrease over 12 days with combination regimens including isoniazid, rifampin, pyrazinamide, ethambutol, and streptomycin, with differences between drugs and drug doses.
Jindani et al, 2003 [62] Prospective cohort	100 newly diagnosed, previously untreated PWPTB; Africa	In the first 2 days, isoniazid has the greatest EBA followed by rifampin, whereas from days 2 to 14, rifampin has significant effects, highlighting its important sterilizing activity. Increased sterilizing effect of pyrazinamide typically occurs after 14 days. Ethambutol may antagonize the sterilizing activities of other drugs although clinical studies have not indicated lower rates of sputum culture conversion.
Cough aerosol sampling systems show decline in culturable <i>Mtb</i> from aerosols after treatment initiation		
Fennelly et al, 2004 [12] Prospective cohort	16 PWPTB (smear-positive) US	Culturable <i>Mtb</i> from cough aerosols decreased rapidly within the first 3 weeks of TB treatment, and cough aerosol culture positivity was associated with lack of TB treatment during the prior week.
Fennelly et al, 2012 [13] Prospective cohort	101 PWPTB (culture-positive); Uganda	Patients demonstrating cough aerosol positivity had received fewer days of TB treatment: 3 vs 4 days for people with non-MDR-TB and 3 vs 5 days for people with MDR-TB ($P = .047$).
Acuña-Villaorduña et al, 2019 [17] Aggregate analysis	233 PWPTB (culture-positive); Uganda and Brazil	Shorter duration of TB treatment before CASS was associated with aerosol positivity: OR, 1.47 per day of treatment (95% CI, 1.16–1.89); $P = .001$. Compared to those who received >2 days of treatment, OR for aerosol positivity was 2.56 (95% CI, 1.11–5.92; $P = .03$) for those who received 1–2 days of treatment, and 5.5 (95% CI, 2.3–13.4; $P < .001$) for those who had not received treatment.
Theron et al, 2020 [18] Cross-sectional	500 PWPTB (including 227 with DR-TB); South Africa	31% patients demonstrated CASS positivity. CASS positivity was more frequent in patients on treatment for shorter durations: 53% of patients at ≤48 h vs 26% on treatment for >48 h ($P < .001$). Culturable <i>Mtb</i> from CASS was detectable in a minority of patients on treatment for 3, 4, 5, and 6 days prior to CASS: 9/26 (35%), 11/31 (35%), 12/37 (32%), and 7/26 (27%), respectively. Drug effectiveness was not associated with the detection of culturable <i>Mtb</i> from CASS in patients on DS-TB regimens or fluoroquinolone-based regimens.
Transcriptomic studies show <i>Mtb</i> gene expression changes that occur after treatment initiation		
Walter et al, 2015 [63] Longitudinal cohort	17 PWPTB (smear-positive, culture-positive DS-TB); Uganda	Bacterial mRNA declined by >98% within 4 days, demonstrating rapid killing, after which rate of decline slowed by >94%.
Shaik et al, 2021 [64] Biological	19 PWPTB (previously untreated): investigation cohort: 7 DS-TB and 2 DR-TB, validation cohort: 10 DS-TB; India	Distinct changes occurred in overall transcriptional profiles when compared pre- and posttreatment (up to 14 days, during which time all remained culture positive), but only in people who received effective treatment. Following exposure to sterilizing TB drugs, <i>Mtb</i> bacilli exhibit transcriptomic changes that may impact pathogenicity and virulence that occur as early as 1 d after treatment initiation.
Walter et al, 2021 [65] Biological	PWPTB from three cohorts (DS-TB, including PWH), Vietnam, Uganda, Benin	Sterilizing anti-TB drugs and highly effective drug regimens (BPaL, BPaMZ) have a rapid and profound effect on suppression of <i>Mtb</i> rRNA synthesis ratio, although CFUs did not decrease substantially until 8–12 d.
Human-to-guinea pig transmission studies show rapid impact of effective treatment on transmission		
Riley, 1957 [66]; Riley et al, 1961 [67] H-GP transmission	61 untreated PWPTB (DS-TB); US	Patients admitted at the time of treatment initiation were 98% less likely to transmit, compared to untreated patients. 14/20 guinea pigs exposed to DR-TB patients were linked to untreated patients; 1 untreated TB patient infected 15 guinea pigs in 3 days.
Escombe et al, 2008 [68] H-GP transmission	97 PWPTB (PWH); Peru	90% of transmission occurred from patients with inadequately treated (undiagnosed) DR-TB rather than those with DS-TB.
Dharmadhikari et al, 2014 [69] H-GP transmission	109 PWPTB (MDR-TB, majority smear-positive) in a series of 5 experimental studies; South Africa	In one of the experiments during which guinea pigs were exposed to 27 patients with MDR-TB newly started on effective treatment, only 1% of guinea pigs developed TB infection. In the other experiments with greater transmission (10%–77% guinea pigs infected), guinea pigs were exposed to patients with unsuspected XDR-TB.
Nardell et al, 2019 [70] H-GP transmission	14 PWPTB (DR-TB, smear-positive); South Africa	0 guinea pigs were infected after exposure to patients with DR-TB who had 72 hours of BPaL (44% infected pretreatment), in contrast to patients who were receiving bedaquiline-containing DR-TB regimens (27% and 28% of guinea pigs infected pre- and posttreatment).
Clinical studies		
Kamat et al, 1966 [80] RCT	191 PWPTB, 693 close family contacts; India	Patients were randomized to receiving treatment entirely at home versus in a sanatorium. At one year, 26% of contacts of patients treated at home versus 23% of contacts of patients treated in the sanatorium developed TB infection (TST conversion).

Table 3. Continued

Study Type and Year	Population and Setting	Key Findings
Brooks et al, 1973 [72] Prospective cohort	21 PWPTB (90% were smear-positive, 76% with cavitory disease), 107 close contacts; US	At 5 years, of contacts who were initially TST negative, 10.5% of home contacts versus 11.5% of sanatorium contacts, developed TB disease, and of contacts who were initially TST positive, 9.4% of home contacts and 15.8% of sanatorium contacts developed TB disease.
Gunnels et al, 1974 [73] Retrospective cohort	Group A: 86 PWPTB (culture-positive), 284 close contacts; Group B: 69 PWPTB (smear- and culture-negative), 216 close contacts; US	No difference in infection (measured by TST conversion) between contacts of Group A patients who were discharged home when culture positive (65% smear-positive) vs contacts of Group B who were smear and culture negative at discharge (40% vs 38%, $P = .54$).
Calderwood et al, 2021 [71] Meta-analysis	44 studies; Various	Smear conversion occurred in 9% of patients at 2 weeks (95% CI, 3%–24%; 1 study), and 82% of patients at 2 months of treatment (95% CI, 78%–86%; 10 studies). For baseline smear-positive patients, culture conversion occurred by 2 weeks in 5% (95% CI, 0%–14%, $n = 2$), increasing to 88% at 2 months (95% CI, 84%–92%, $n = 20$).

Abbreviations: BPaL, bedaquiline, pretomanid, linezolid; BPaMZ, bedaquiline, pretomanid, moxifloxacin, pyrazinamide; CASS, cough aerosol sampling system; CFU, colony-forming units; CI, confidence interval; DR-TB, drug-resistant tuberculosis; DS-TB, drug-susceptible tuberculosis; EBA, early bactericidal activity; H-GP, human-to-guinea pig; MDR-TB, multidrug-resistant tuberculosis; mRNA, messenger RNA; *Mtb*, *Mycobacterium tuberculosis*; PWPTB, people with pulmonary tuberculosis; PWH, people with human immunodeficiency virus; OR, odds ratio; RCT, randomized controlled trial; rRNA, ribosomal RNA; TB, tuberculosis; TST, tuberculin skin test; US, United States; XDR-TB, extensively drug-resistant tuberculosis.

Experimental Human-to-Guinea Pig Transmission Studies Consistently Demonstrate the Rapid Effect of TB Treatment on Transmission

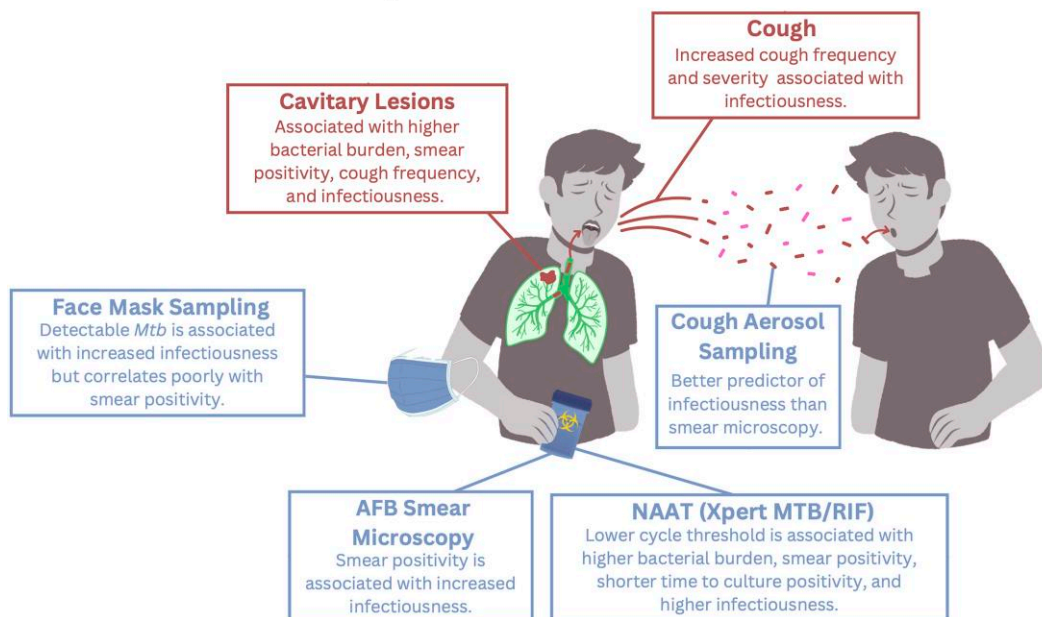
Experimental studies in the United States, Peru, and South Africa used specially constructed wards that enable air exhaled by PWPTB to be directed to susceptible guinea pigs measure the effect of interventions, like treatment, on transmission, measured directly based on GP infection (TST conversion) [66, 68, 69, 79]. H-GP studies demonstrate wide variability in infectiousness among untreated patients, and that transmission appears to cease rapidly (within days) after effective treatment initiation, irrespective of sputum bacteriologic status [66, 67, 69]. Early H-GP studies demonstrated that patients with pulmonary DS-TB admitted to the experimental ward at the time of initiating treatment with isoniazid, streptomycin, and para-aminosalicylic acid were 98% less likely to transmit, compared to untreated patients, suggesting an almost immediate effect [67]. Another H-GP study on a ward in Lima, Peru, demonstrated that 90% of transmission arose from patients with TB/HIV coinfection with inadequately treated (undiagnosed) DR-TB rather than those with DS-TB [79]. In a series of retrospective analyses of H-GP transmission studies in South Africa, after 3 months of exposure to 27 patients with multidrug-resistant TB, who were initiated on effective treatment at the time they entered the experimental ward, only 1% of guinea pigs developed infection, in contrast to other experiments in which there were patients with undiagnosed and untreated extensively drug-resistant TB (10%–77% infected guinea pigs) [69]. More recent unpublished H-GP studies in South Africa examined the effect of the BPaL (bedaquiline, pretomanid, linezolid) regimen and found that zero guinea pigs were infected after exposure to patients with DR-TB who had received BPaL for 72 hours, in contrast to patients who were

receiving prior DR-TB regimens including bedaquiline and linezolid, in which no difference in guinea pig infections was seen (28% vs 26%, $P = 1.00$) [70]. These data suggest a rapid decrease in infectiousness within days of BPaL initiation and further suggest that certain drugs in the regimen such as pretomanid may have a differential effect on infectiousness [70].

Clinical Studies Support a Rapid Decline in Infectiousness and Indicate That Smear Status Is a Poor Predictor of Transmission Risk After Treatment Initiation

Data from a randomized controlled trial in India in the 1950s demonstrated similar rates of household transmission, regardless of index patient smear status, when comparing patients initiated on anti-TB treatment (ATT) at home (ie, ongoing [re]-exposure for household contacts) versus hospitalized in a sanatorium for 12 months (26% versus 23% contact TST conversion at 1 year, 10.5% versus 11.5% contact TB disease at 5 years (in baseline TST negative contacts)) [80–82]. This suggests a rapid effect of ATT without increased incremental household risk from PWPTB despite ongoing exposure. A prospective study of 21 patients with TB discharged from Cincinnati General Hospital after 2 weeks of ATT (90% remained smear-positive, 76% had cavitory disease) found that 97% of contacts (72/74) who were TST negative at baseline did not convert [72]. A retrospective cohort study from Arkansas showed no difference in TB infection in contacts exposed to index patients who were smear-positive versus smear-negative at the time of hospital discharge (TST >9 mm: 40% vs 38%) [73]. While PWPTB in these studies were on treatment for 2 weeks prior to discharge home, and these studies were non-randomized, which could lead to selection bias, these data suggest that infection risk after ATT initiation is low and

Pretreatment: Tests and clinical factors that predict infectiousness



Posttreatment: Effect of treatment on infectiousness

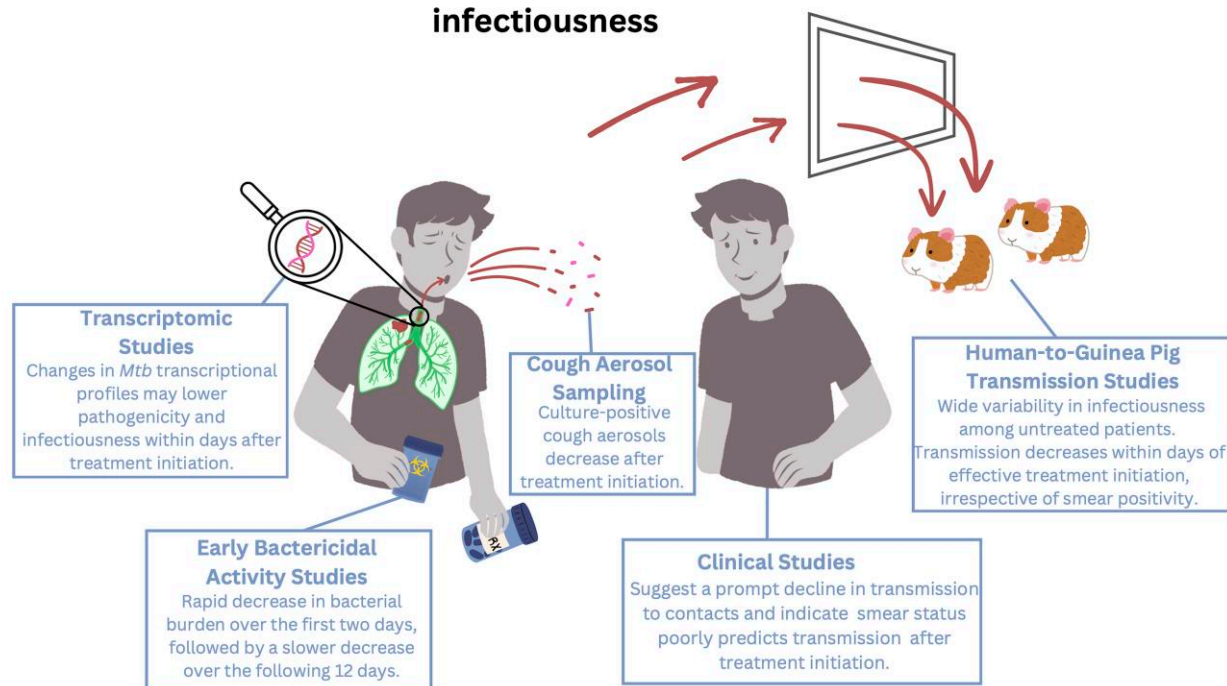


Figure 1. Existing and novel tools and approaches to measure infectiousness pre- and post-treatment for pulmonary tuberculosis. Abbreviations: AFB, acid-fast bacilli; *Mtb*, *Mycobacterium tuberculosis* complex; NAAT, nucleic acid amplification test.

persistent smear positivity after ATT initiation does not increase contact TB infection [72, 73, 83]. This is important since smear and culture conversion often happen after 2 weeks [71].

Although smear and culture conversion are widely used to assess treatment response, available data do not support these being predictors of infectiousness in people on effective ATT [84].

CONCLUSIONS (FIGURE 1)

There is heterogeneity in infectiousness among PWPTB. Prior to ATT, positive sputum smear, lower Xpert Ct, and/or presence of cough and cavities predict infectiousness risk. New tools like CASS and FMS may predict transmission better than sputum-based tests, but are not clinically available. The exact duration of infectiousness after treatment initiation remains uncertain and likely varies between individuals. Sputum smear and culture status are poor predictors of infectiousness after treatment. Data from EBA studies, experimental H-GP studies, clinical studies, and transcriptomic studies suggest a rapid and steady impact of ATT (within 2–4 days) and indicate that persistent bacteriologic positivity does not necessarily represent ongoing infectiousness in patients on effective treatment. Limitations of the data reviewed include heterogeneous study designs, small cohorts with limited reproducibility, lack of head-to-head comparisons of different tools to assess infectiousness, and limited experimental models that reliably inform infectiousness in human populations in real-world settings.

Further studies are needed to test individual drugs and drug combinations for their ability to stop transmission under controlled clinical conditions at different time periods post-treatment initiation, correlated with *Mtb* transcriptomic evaluation. Combining data from CASS, FMS, RASC, and approaches such as the H-GP model as part of household, community, and facility transmission studies can advance understanding of infectiousness and ultimately inform clinical and public health decisions. In the meantime, existing and emerging data can inform more nuanced evidence-based isolation recommendations that are better aligned with person-centered care and public health goals.

Notes

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